

//20 MCQs on loop conditions

1. What is the purpose of a loop condition in C? A) To initialize variables B) To increment variables C) To control loop execution D) To declare variables

Answer: C) To control loop execution

2. Which loop condition is evaluated before the loop body is executed? A) for loop B) while loop C) do-while loop D) Both A and B

Answer: D) Both A and B

3. What is the syntax for a for loop condition in C? A) for (init; condition; increment) B) for (condition; init; increment) C) for (init; increment; condition) D) for (increment; init; condition)

Answer: A) for (init; condition; increment)

4. What happens if the loop condition is false initially? A) The loop body is executed once B) The loop body is skipped C) The loop body is executed infinitely D) The program crashes

Answer: B) The loop body is skipped

5. Which loop is guaranteed to execute at least once? A) for loop B) while loop C) do-while loop D) None of the above

Answer: C) do-while loop

6. What is the loop condition for an infinite loop in C? A) 1 B) 0 C) true D) Both A and C

Answer: D) Both A and C

7. How do you exit a loop prematurely in C? A) Using break statement B) Using continue statement C) Using return statement D) Using exit statement

Answer: A) Using break statement

8. What is the purpose of the continue statement in C? A) To exit the loop B) To skip the current iteration C) To restart the loop D) To pause the loop

Answer: B) To skip the current iteration

9. Which loop is best suited for iterating over an array in C? A) for loop B) while loop C) do-while loop D) None of the above

Answer: A) for loop

10. What is the condition for a while loop in C? A) while (condition) B) while (!condition) C) while (true) D) while (false)

Answer: A) while (condition)

11. What happens if the loop condition is missing in a for loop? A) The loop is skipped B) The loop is executed once C) The loop is executed infinitely D) The program crashes

Answer: C) The loop is executed infinitely

12. How do you specify multiple conditions in a loop in C? A) Using logical AND (&&) B) Using logical OR (||) C) Using bitwise AND (&) D) Using bitwise OR (|)

Answer: A) Using logical AND (&&) or B) Using logical OR (||)

13. What is the purpose of loop nesting in C? A) To execute loops concurrently B) To execute loops sequentially C) To simplify loop logic D) To improve performance

Answer: B) To execute loops sequentially

14. Which loop is best suited for reading input from a user in C? A) for loop B) while loop C) do-while loop D) None of the above

Answer: C) do-while loop

15. What is the condition for a do-while loop in C? A) do { ... } while (condition); B) do { ... } while (!condition); C) do { ... } while (true); D) do { ... } while (false);

Answer: A) do { ... } while (condition);

16. How do you iterate over a linked list in C? A) Using a for loop B) Using a while loop C) Using a do-while loop D) Using recursion

Answer: B) Using a while loop

17. What is the purpose of loop unrolling in C? A) To reduce loop overhead B) To increase loop overhead C) To simplify loop logic D) To improve performance

Answer: A) To reduce loop overhead or D) To improve performance

18. Which loop is best suited for implementing a timer in C? A) for loop B) while loop C) do-while loop D) None of the above

Answer: B) while loop

19. What is the condition for an empty loop in C? A) ; B) {} C) () D) []

Answer: A) ;

20. How do you optimize loop performance in C? A) By reducing loop iterations B) By increasing loop iterations C) By using loop unrolling D) By using loop fusion

Answer: A) By reducing loop iterations or C) By using loop unrolling